INDIAN INSTITUE OF TECHNOLOGY MADRAS

LIBRARY MANAGEMENT SYSTEM

Modern Application Development 1 Project

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Introduction:

This is a multi-user application for a Library Management System which is meant to issue books to users.

Methodology:

Tools and technologies

Tools Used:

Flask for main application

Flask-SQLAlchemy for database connection

Flask-Login for login system

Flask-Restful for API

Apscheduler for scheduling background task (automatic return of book)

Matplotlib.pyplot for graphs on stats page (numpy as well)

Base64 for encoding graphs into base64 to send in api response body

Architecture:

The client accesses the app from a web browser like Edge, Firefox, Chrome, Brave, etc.

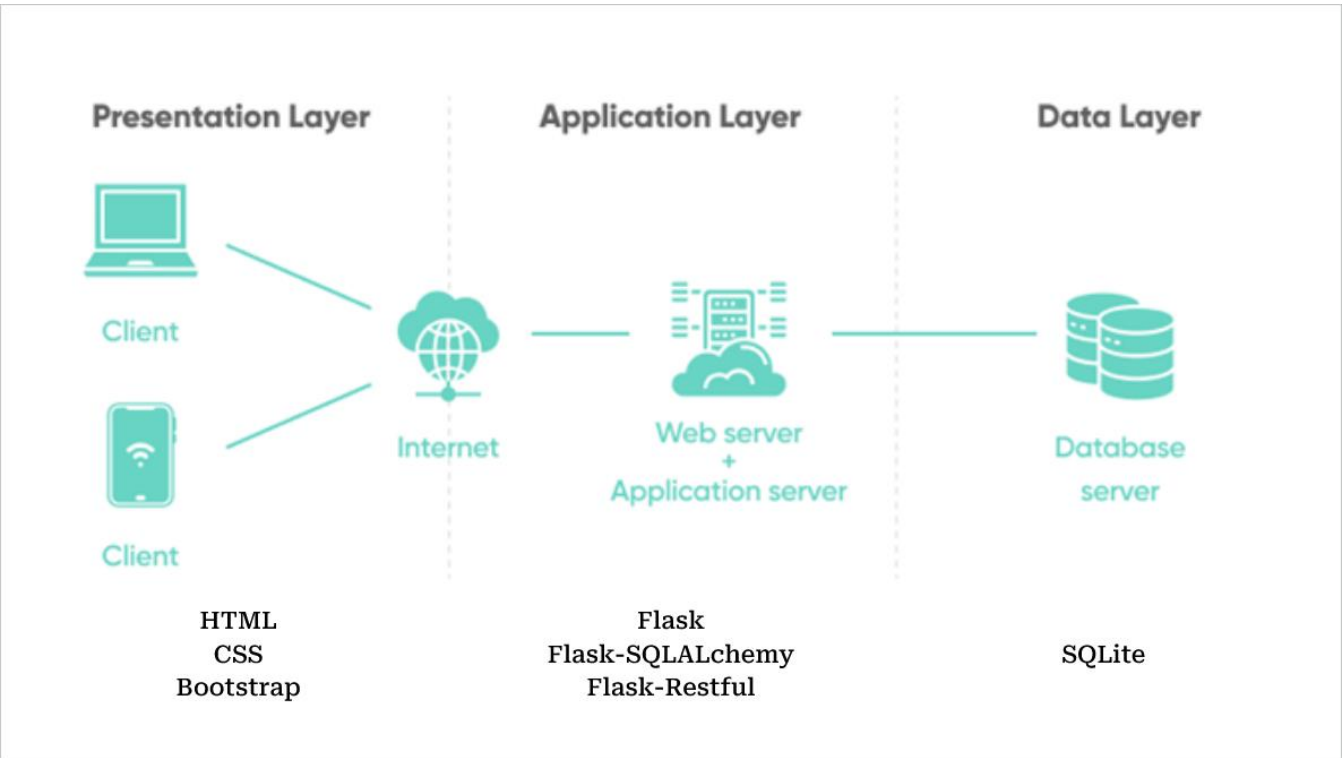
The webpage is made using HTML, CSS and Bootstrap (and little JavaScript)

The application is run using Flask

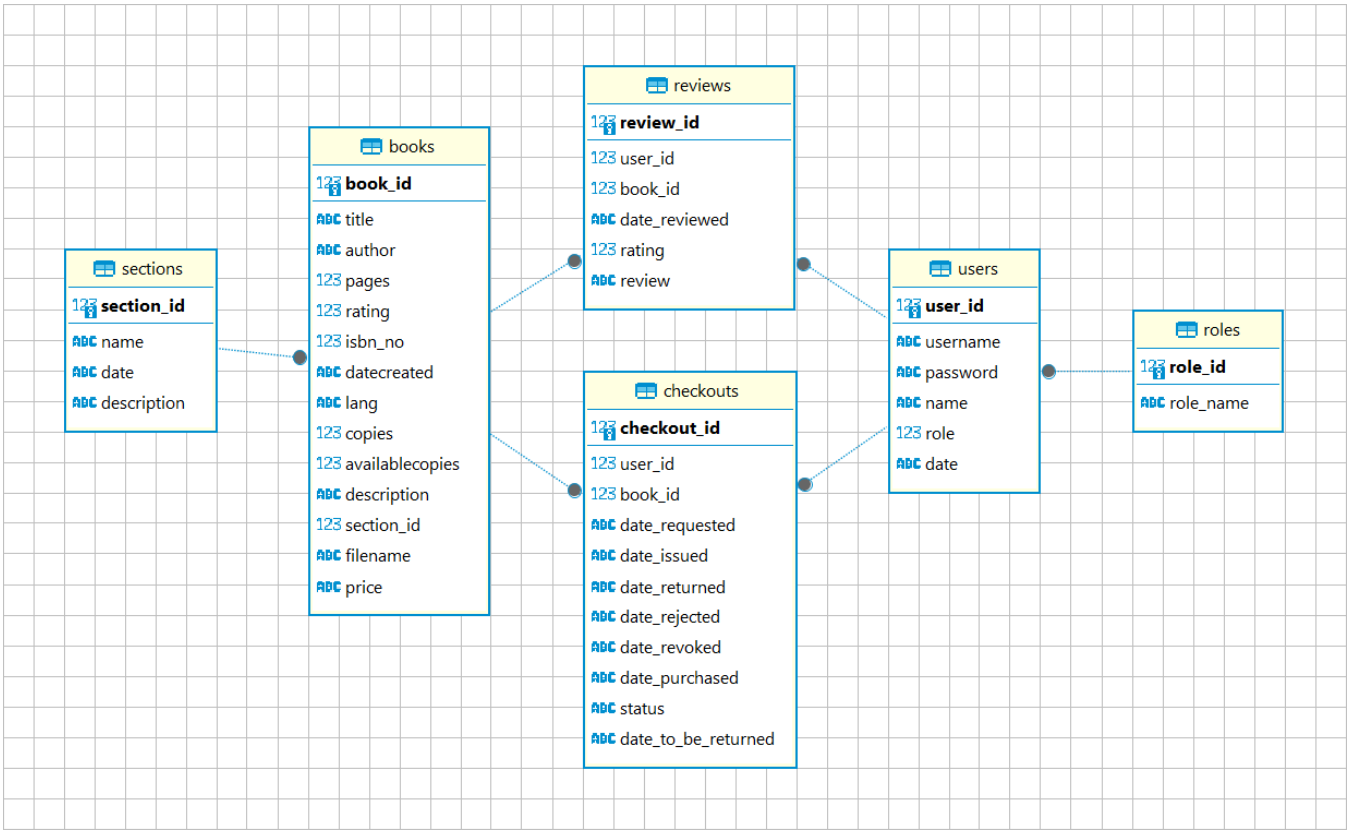
The database connection is made using Flask SQLAlchemy

The APIs are created using Flask Restful

The data is stored in SQLite database



Database Schema:



Implementation:

I started by making sample webpages using minimal bootstrap for a few of the pages to get an idea, and connecting them to the endpoints. Then I created simple database models for sections and books, focusing on setting up the library's structure. I then implemented the functionality for adding and editing books and sections, along with cascading deletion for sections.

I deliberated over whether to have separate tables for book catalogue and book details, opting to begin with a single table. To address the issue of checking if there were sufficient books, two attributes were introduced: available copies and total copies, ensuring availability when issuing books.

I then created the user table and linked it with the login page (without a login framework), and allowed users to request books. I also created the stats page using matplotlip and numpy.

After looking at other websites, I changed a few things like naming convention for endpoints and navbar options.

After identifying recurring code patterns, I created templates for various components, such as the navbar, search bar, book card, and section card.

I then created APIs for CRUD on books and sections

I felt it difficult to do the automatic renewal, so I added a new column named date_to_be_returned which is calculated and updated when the book is being issued. This attribute isn't accessible to anyone so there is no way for the user to update/change it. Using apscheduler I added a job to check whether the date to be returned had passed, and if it has, returns the book.

To make a book available for download when a user purchases it, I created a new endpoint and used send_file to download on click.

At the end, I added extra functionality like API's for sending graphs (by encoding in base64), custom error pages for 401, 404 and 500, and profile settings like changing user's name and password.

Conclusion:

Lessons learnt:

Changing the database model from the original to meet new requirements

Creating templates/functions for repetitive code

Use the same URLs for user and librarian (instead of user_dash and librarian_dash, create one dashboard and different pages rendered based on current user)

Necessary to verify the datatypes in the frontend itself

Thigs I Would Implement in the Future:

Would find a different way to store the date returned/revoked/rejected instead of storing them all in their own columns

Would create a book catalogue table with accession number as primary key

Find a different way to store pdfs: either in database itself or zipping it

Instead of disabling toolbar, I would find another way to remove download option

Would format the page so that it is reactive to change in screen size (mobile, tablet, etc.)

Create a new table for authors which has many-many relationship with books

Add more attributes to a user such as age, gender, etc. to create new graphs based on this information. This could be preferred genre by age/gender, or number of books read by each age group.

Would use JavaScript to make a better User Experience

Video Link:

https://drive.google.com/file/d/1r_N7zJ4o3baeBPCAiUBLBjmBSkm1Oh0p/view?usp=sharing